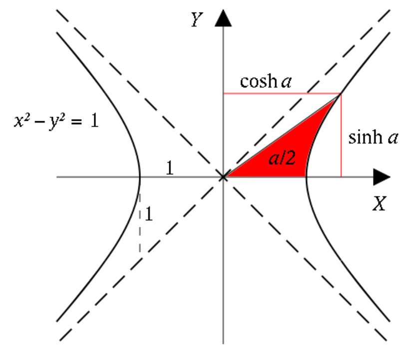
1.8 Hyperbolic Trigonometric Functions

Motivation: The trigonometric functions work well when we are trying to do trigonometry on a plane (intuitively a plane is a flat, infinite two dimensional object).

However, it doesn't always make sense to do trigonometry on the plane. For example, the earth is roughly a sphere and trigonometry on a sphere is different than trigonometry on a plane. Another example where trigonometry will be different is if we are on a hyperbolic plane.



This leads to the following definitions.

Definitions: We define six <u>hyperbolic trig functions</u>.

Pronounced "sinch" $\sinh(x) = \dfrac{e^x - e^{-x}}{2}$ $\qquad$ $\operatorname{csch}(x) = \dfrac{1}{\sinh(x)}$

Pronounced "cosh" $\cosh(x) = \dfrac{e^x + e^{-x}}{2}$ $\qquad$ $\operatorname{sech}(x) = \dfrac{1}{\cosh(x)}$

$\tanh(x) = \dfrac{\sinh(x)}{\cosh(x)}$ $\qquad$ $\coth(x) = \dfrac{1}{\tanh(x)}$

Example 1: Evaluate each of the following using the definition.

a) $\sinh(0)$

b) $\cosh(1)$

c) $\tanh(2)$

Graphs of hyperbolic functions:

$y = \sinh(x)$ $\qquad$ $y = \cosh(x)$ $\qquad$ $y = \tanh(x)$

Theorem: We have the following identities for all real x that are in the domain of the functions involved in each identity:

1) $\sinh(-x) = -\sinh(x)$
2) $\cosh(-x) = \cosh(x)$
3) $\cosh^2(x) - \sinh^2(x) = 1$
4) $1 - \tanh^2(x) = \operatorname{sech}^2(x)$
5) $\coth^2(x) - 1 = \operatorname{csch}^2(x)$

Example 2: Show that $\cosh^2(x) - \sinh^2(x) = 1$.

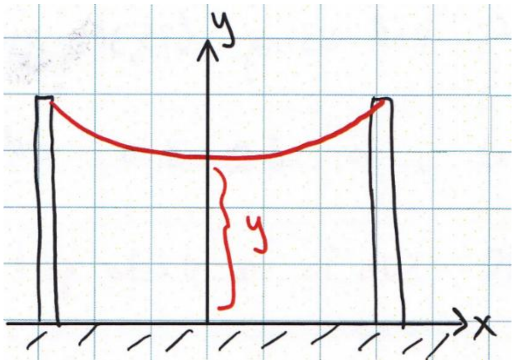
Example 3: Find $\frac{d}{dx}\left[\sinh(x)\right]$ and $\frac{d}{dx}\left[\cosh(x)\right]$.

Theorem: We have the following derivatives for our six hyperbolic trig. functions.

- $\frac{d}{dx}\left[\sinh(x)\right] =$
- $\frac{d}{dx}\left[\cosh(x)\right] =$
- $\frac{d}{dx}\left[\tanh(x)\right] =$
- $\frac{d}{dx}\left[\coth(x)\right] =$
- $\frac{d}{dx}\left[\operatorname{sech}(x)\right] =$
- $\frac{d}{dx}\left[\operatorname{csch}(x)\right] =$

Applications
Hyperbolic functions arise when a homogeneous cable is suspended between two points.

Catenary Curve



$$y = a\cosh\left(\frac{x}{a}\right) + C$$

(where *a* and *c* are constants driven by the type of cable and the distance between the poles.)

Example 4: Suppose a cable is attached between two poles which stand 30ft apart and can be modelled by the function $y = \cosh(x) + 5$. Determine the exact length of the cable from $0 \le x \le 10$.

Example 5: Consider the region bounded by the curves $y = \sinh(x)$ & $y = -x^2 - 1$ from $0 \le x \le 1$.

a) Determine the exact value of the area of this region.

b) Set up, *but do not evaluate*, and integral which represents the volume when this region is revolved around the y-axis.

c) Suppose you rotate the curve $y = \sinh(x)$ on $-1 \le x \le 1$ about the x-axis and attempt to determine the surface area of the resulting solid. Why does the integral $\int_{-1}^{1} 2\pi \sinh(x)\sqrt{1+\cosh^2(x)}\,dx$ *not* represent this surface area?